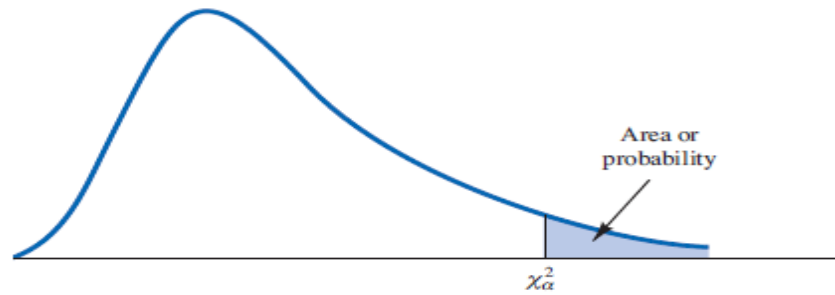


χ^2 table



Degrees of Freedom	Area in Upper Tail							
	.99	.975	.95	.90	.10	.05	.025	.01
1	.000	.001	.004	.016	2.706	3.841	5.024	6.635
2	.020	.051	.103	.211	4.605	5.991	7.378	9.210
3	.115	.216	.352	.584	6.251	7.815	9.348	11.345
4	.297	.484	.711	1.064	7.779	9.488	11.143	13.277
5	.554	.831	1.145	1.610	9.236	11.070	12.832	15.086
6	.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.041	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.878	14.573	16.151	18.114	36.741	40.113	43.195	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892
40	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691
60	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379
80	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329
100	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807

**Note: A more extensive table is provided as Table 3 of Appendix B.*

Chapter Four

Chi-Square Distributions (χ^2)

4.1. Areas of application

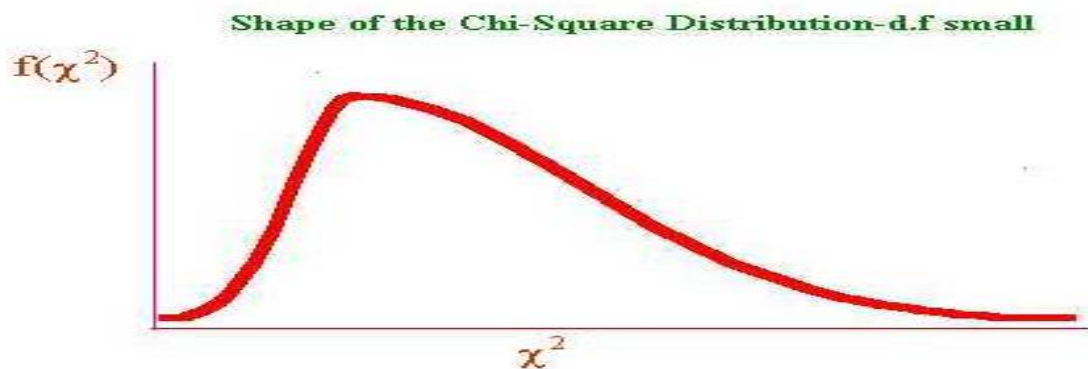
In previous chapters, we have based our decisions on the assumption that the population was normally distributed. When this assumption about the population cannot be made, then it becomes necessary to use other procedures. One of the tests used in such situations is known as *Chi square (X^2) test*. This test is good for nominal or ordinal scale of measurement.

X^2 distribution has the following characteristics

1. It involves squared observations and hence it is always positive.
2. The distribution is skewed to the right so that its skewness is positive.
3. Similar to *t distribution*, there is a family of chi-square distributions. There is a particular distribution for each degree of freedom.

The degree of freedom (df) are calculated as follows;

- a. $df = (k-1)$, where K is the number of categories in one sample test
- b. $df = (K-1)(r-1)$, where K is the number of columns and r is the number of rows in a cross classification table (contingency table) for various categories of two or more independent samples.



4.2. Tests for independence between two variables

The chi-square **independence test** can be used to test the independence of two variables.

Steps:

1. State the null and alternative hypothesis

H_0 : the column variable is independent of the row variable

H_1 : the column variable is not independent of the row variable

2. Select a random sample and record the observed frequencies
3. Compute expected frequencies
4. Compute the value of the test statistic
5. Decision rule: Reject H_0 if $X^2 \geq X^2_{\alpha}$

Illustration: Let us illustrate the test of independence by considering the study conducted by the MOHA soft drink company. MOHA manufactures and distributes three types of Soft drink: Mirinda, 7 UP, and Pepsi. In an analysis of the market segments for the three Soft drinks, the firm's market research group raised the question of whether preferences for the three Soft drinks differ among male and female drinkers. If soft drinks preference is independent of the gender of the drinker, one advertising campaign will be initiated for all of MOHA's soft drink. However, if soft drink preference depends on the gender of the drinker, the firm will tailor its promotions to different target markets.

Step 1:

H_0 : Soft drink preference is independent of the gender of the drinker.

H_1 : Soft drink preference is not independent of the gender of the drinker.

Step 2: After identification of the population as all male and female drinkers, a sample selected and each individual asked to state his or her preference for the three soft drinks.

Contingency table for soft drink preference and gender*

GENDER	SOFT DRINK PREFERENCE			
	Mirinda	7 UP	Pepsi	Total
Male	20	40	20	80
Female	30	30	10	70
Total	50	70	30	150

*When data are arranged in table form for the chi-square test, the table is called a **contingency table**.

Step 3: Expected Frequencies

GENDER	SOFT DRINK PREFERENCE			
	Mirinda	7 UP	Pepsi	Total
Male	26.67	37.33	16	80
Female	23.33	32.67	14	70
Total	50	70	30	150

Expected frequency = $\frac{(\text{Row total})(\text{Column total})}{\text{Sample size}}$

TEST STATISTIC FOR INDEPENDENCE

$$\chi^2 = \sum_i \sum_j \frac{(f_{ij} - e_{ij})^2}{e_{ij}}$$

Step 4:

where

f_{ij} = observed frequency for contingency table category in row i and column j

e_{ij} = expected frequency for contingency table category in row i and column j
based on the assumption of independence

Computation of the chi square test statistic

Gender	Preference	Observed frequency(f_{ij})	Expected frequency(e_{ij})	$f_{ij} - e_{ij}$	$(f_{ij} - e_{ij})^2$	$\frac{(f_{ij} - e_{ij})^2}{e_{ij}}$
Male	Mirinda	20	26.67	-6.67	44.44	1.67
Male	7 UP	40	37.33	2.67	7.11	0.19
Male	Pepsi	20	16	4.00	16.00	1.00
Female	Mirinda	30	23.33	6.67	44.44	1.90
Female	7 UP	30	32.67	-2.67	7.11	0.22
Female	Pepsi	10	14	4.00	16	1.14
	Total	150				$X^2=6.12$

Step 5: Degree of freedom= (2-1)(3-1)= 2

$X^2_{0.05}$ at degree of freedom=2 is 5.991, $X^2 > X^2_{\alpha}$, therefore Reject H_0 . The preference is not independent of the gender of the drinker.

4.3. Tests for the equality of several proportions

In this situation, samples are selected from several different populations, and the researcher is interested in determining whether the proportions of elements that have a common characteristic are the same for each population.

Illustration: A psychologist selected 100 people from each of four income groups and asked them if they were “very happy.” The percent for each group who responded yes and the number from the survey are shown in the table. At a 0.05 level of significance test the claim that there is no difference in the proportions.

Household income	Less than 30,000 (24%)	30,000 – 74,999 (33%)	75,000 – 99,999 (38%)	100,000 or more (49%)	Total
Yes	24	33	38	49	144
No	76	67	62	51	256
Total	100	100	100	100	400

Solution

Step 1: State the hypotheses and identify the claim.

$$H_0: p_1 = p_2 = p_3 = p_4$$

H_1 : At least one proportion differs from the others.

Step 2: Find the critical value. $(R - 1)(C - 1) = (2 - 1)(4 - 1) = 3$. The critical value is 7.815.

Step 3: Compute the test value.

EXPECTED FREQUENCIES

Household income	Less than 30,000 (24%)	30,000 – 74,999 (33%)	75,000 – 99,999 (38%)	100,000 or more (49%)	Total
Yes	36	36	36	36	144
No	64	64	64	64	256
Total	100	100	100	100	400

Calculation of χ^2

$$\begin{aligned}
 \chi^2 &= \sum \frac{(O - E)^2}{E} \\
 &= \frac{(24 - 36)^2}{36} + \frac{(33 - 36)^2}{36} + \frac{(38 - 36)^2}{36} + \frac{(49 - 36)^2}{36} \\
 &\quad + \frac{(76 - 64)^2}{64} + \frac{(67 - 64)^2}{64} + \frac{(62 - 62)^2}{64} + \frac{(51 - 64)^2}{64} \\
 &= 4 + 0.25 + 0.1111 + 4.6944 + 2.25 + 0.1406 + 0.0625 + 2.6406 \\
 &= 14.149
 \end{aligned}$$

Step 4: Make the decision reject the null hypothesis since $14.149 > 7.815$.

4.4. Goodness- of fit tests

The X^2 test is used to test whether there is a significance difference between the observed number of responses in each category and the expected number of responses for such category under the assumptions of null hypothesis. In other words, the objective is to find out how well the distribution of observed frequencies (f_o) fit the distribution of expected frequencies (f_e).

Illustration: Assume that a shoe manufacturer company sold 30 pairs of shoes. There is six size of shoes and the company expects the size of shoes are sold in equal proportion that is each size is sold 5 times. In the survey conducted, the actual number of frequency each size sold in sequence is as follows; 4, 7, 3, 6, 8, 2. A comparison of observed frequency and expected frequency is tabulated below;

	Size					
	A	B	C	D	E	F
Observed frequency	4	7	3	6	8	2
Expected frequency	5	5	5	5	5	5

Steps involved in X^2 test

Step 1: H_0 : All sizes are equally likely to occur.

$$P_1=P_2=P_3=P_4=P_5=P_6$$

H_1 : All probabilities are not equally likely to occur.

Step 2: Assume $\alpha = 0.05$.

Step 3: Expected frequency (f_e) = 5.

Step 4:

$$\begin{aligned} X^2 &= (4-5)^2/5 + (7-5)^2/5 + (3-5)^2/5 + (6-5)^2/5 + (8-5)^2/5 + (2-5)^2/5 \\ &= 1/5 + 4/5 + 4/5 + 1/5 + 9/5 + 9/5 \\ &= 28/5 = \underline{5.6} \end{aligned}$$

Step 5: Critical value of $X^2_{0.05}$ and $df = (K-1) = (6-1) = 5$ is given as 11.07.

Since our computed value of $X^2 = 5.6$ is less than the critical value of $X^2=11.07$, we can accept the null hypothesis.